PP-Module for LiFi Access Systems

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National Information Assurance Partnership

Revision History

Version	Date	Comment
1.0	2026-02-24	Initial Release

Contents

1 Introduction

1.1 Overview

The scope of the LiFi Access System PP-Module is to describe the security functionality of a LiFi access system product in terms of [CC] and to define functional and assurance requirements for such products. This PP-Module is intended for use with the following Base-PPs:

- collaborative Protection Profile for Network Devices, Version 4.0

A conformant TOE will claim conformance to a PP-Configuration that includes the Base-PP identified above, this PP-Module, and any of the PP-Modules or packages identified in Section 2.

This Base-PP is valid because a LiFi Access System is a specific function for a purpose-built network device.

1.2 Terms

The following sections list Common Criteria and technology terms used in this document.

1.2.1 Common Criteria Terms

Assurance	Grounds for confidence that a TOE meets the SFRs [CC].
Base Protection Profile (Base-PP)	Protection Profile used as a basis to build a PP-Configuration.
Collaborative Protection Profile (cPP)	A Protection Profile developed by international technical communities and approved by multiple schemes.
Common Criteria (CC)	Common Criteria for Information Technology Security Evaluation (International Standard ISO/IEC 15408).
Common Criteria Testing Laboratory	Within the context of the Common Criteria Evaluation and Validation Scheme (CCEVS), an IT security evaluation facility accredited by the National Voluntary Laboratory Accreditation Program (NVLAP) and approved by the NIAP Validation Body to conduct Common Criteria-based evaluations.
Common Evaluation Methodology (CEM)	Common Evaluation Methodology for Information Technology Security Evaluation.
Distributed TOE	A TOE composed of multiple components operating as a logical whole.
Operational Environment (OE)	Hardware and software that are outside the TOE boundary that support the TOE functionality and security policy.
Protection Profile (PP)	An implementation-independent set of security requirements for a category of products.

Protection Profile Configuration (PP-Configuration)	A comprehensive set of security requirements for a product type that consists of at least one Base-PP and at least one PP-Module.
Protection Profile Module (PP-Module)	An implementation-independent statement of security needs for a TOE type complementary to one or more Base-PPs.
Security Assurance Requirement (SAR)	A requirement to assure the security of the TOE.
Security Functional Requirement (SFR)	A requirement for security enforcement by the TOE.
Security Target (ST)	A set of implementation-dependent security requirements for a specific product.
Target of Evaluation (TOE)	The product under evaluation.
TOE Security Functionality (TSF)	The security functionality of the product under evaluation.
TOE Summary Specification (TSS)	A description of how a TOE satisfies the SFRs in an ST.

1.2.2 Technical Terms

Access Point (AP)	A device that provides the network interface that enables wireless client hosts to access a wired network.
LiFi	A wireless computer network that links two or more devices using modulated optical communication to form a local area network within a limited area such as a home, school, computer laboratory, campus, office building, etc.
Service Set Identifier (SSID)	The primary name associated with an 802.11 wireless local area network (WLAN).

1.3 Compliant Targets of Evaluation

This PP-Module specifically addresses LiFi (IEEE 802.11bb, ITU G.9991) access systems (AS). A compliant LiFi AS is a system composed of hardware and software that is connected to a network and has an infrastructure role in the overall enterprise network. In particular, a LiFi AS establishes a secure wireless (IEEE 802.11, IEEE802.1x, IEEE 802.1AE) link that provides an authenticated and encrypted path to an enterprise network and thereby decreases the risk of exposure of information transiting “over-the-air.”

Since this PP-Module extends the NDcPP, conformant TOEs are obligated to implement the functionality required in the NDcPP along with the additional functionality defined in this PP-Module in response to the threat environment discussed subsequently herein.

The TOE encompasses the OS kernel drivers, shared software libraries, and some application software included with the OS. The applications considered within the TOE are those that provide essential LiFi services that make up the LiFi link and security. Examples are management services that implement configuration functions specified in this PP-Module, host services that manage client authentication to the access point, and the libraries and modules that provide security implementation of the LiFi link. This applies to each separate TOE component that is being used to meet the security requirements proposed.

1.3.1 TOE Boundary

The physical boundary for a TOE that conforms to this PP-Module is one or more physical network appliances (in a standalone or distributed configuration) that provides generalized network device functionality such as auditing, identification and authentication, and cryptographic services for network communications, as well as specialized functionality for facilitating remote network access via LiFi communications. The TOE's logical boundary includes all functionality required by the claimed Base-PP, this PP-Module, and any other PP-Modules that may be claimed by the ST author as a part of a PP-Configuration that includes this PP-Module. Product functionality for which no PP-Module exists, or for which no SFR within a relevant PP-Module exists to capture, are considered to be non-interfering with respect to the TSF. In other words, they may be present or enabled in the evaluated configuration of the TOE but their presence must not degrade or interrupt the enforcement of functions within the TOE's logical boundary.

Note that while the Base-PP permits the TOE to be either a physical or virtual network appliance, LiFi devices require the use of specialized hardware that will not be present on commodity servers, and it is therefore expected that any TOE that conforms to this PP-Module will be one or more dedicated physical devices.

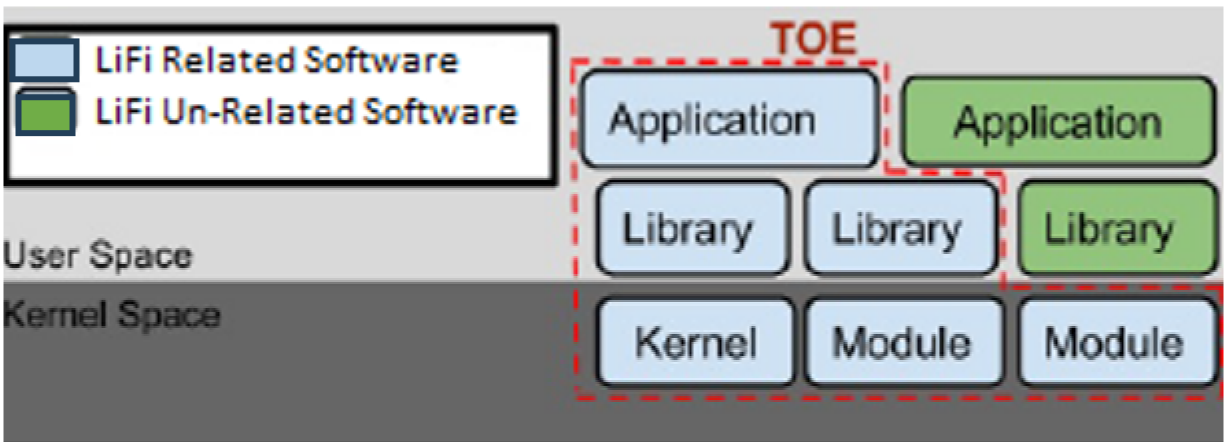


Figure 1: High Level TOE Implementation

1.4 Use-Cases

[USE CASE 1] Standalone Device

The TOE is a standalone network device that serves as a single network endpoint providing connectivity to LiFi clients.

[USE CASE 2] Distributed System

The TOE is a distributed system consisting of multiple network devices that collectively serve as the LiFi network endpoint. In addition to claiming the relevant "Distributed TOE" requirement in the NDcPP, this use case also requires the TOE to claim the optional SFR FCS_CKM.2/DISTRIB to describe the key distribution method between distributed TOE components.

1.5 Implementation-Based Features

A conformant TOE may implement either 802.11bb or G.vlc. Depending on which is supported, different requirements will apply to the TOE as they each implement different methods of protection of data in transit.

1.5.1 802.11bb Support

The TOE implements LiFi using IEEE 802.11bb. Data in transit is protected using either IPsec or RadSec (RADIUS over TLS).

If this feature is implemented by the TOE, the following requirements must be claimed in the ST:

- **FCS_CKM.1/WPA**
- **FCS_CKM.2/GTK**
- **FCS_CKM.2/PMK**
- **FTP_ITC.1/Client**

1.5.2 Distributed TOE

A specific subtype of a TOE that supports 802.11bb is a distributed architecture. When the TOE implements this feature, it uses a key distribution mechanism to pass keys between distributed components so that a remote client may persistently maintain a connection to different connections in transit (e.g., when moving to different floors of a building).

If this feature is implemented by the TOE, the following requirements must be claimed in the ST:

- **FCS_CKM.2/DISTRIB**

1.5.3 G.vlc Support

The TSF implements LiFi using ITU-T G.9991 (aka G.vlc). Data in transit is protected using MACsec. When the TOE implements this feature, it must also claim conformance to the PP-Module for MACsec Ethernet Encryption.

If this feature is implemented by the TOE, the following requirements must be claimed in the ST:

- **FTP_ITC.1/Gvlc**

2 Conformance Claims

Conformance Statement

An ST must claim exact conformance to this PP-Module.

The evaluation methods used for evaluating the TOE are a combination of the workunits defined in [CEM] as well as the Evaluation Activities for ensuring that individual SFRs and SARs have a sufficient level of supporting evidence in the Security Target and guidance documentation and have been sufficiently tested by the laboratory as part of completing ATE_IND.1. Any functional packages this PP claims similarly contain their own Evaluation Activities that are used in this same manner.

CC Conformance Claims

This PP-Module is conformant to Part 2 (extended) and Part 3 (conformant) of Common Criteria CC:2022, Revision 1.

PP Claim

This PP-Module does not claim conformance to any Protection Profile.

The following PPs and PP-Modules are allowed to be specified in a PP-Configuration with this PP-Module:

- collaborative Protection Profile for Network Devices, Version 4.0
- PP-Module for MACsec Ethernet Encryption, version 2.0

Package Claim

- This PP-Module is Functional Package for Transport Layer Security Version 2.1 conformant.
- This PP-Module is Functional Package for X.509 Version 1.0 conformant.
- This PP-Module does not conform to any assurance packages.

The functional packages to which the PP conforms may include SFRs that are not mandatory to claim for the sake of conformance. An ST that claims one or more of these functional packages may include any non-mandatory SFRs that are appropriate to claim based on the capabilities of the TSF and on any triggers for their inclusion based inherently on the SFR selections made.

3 Security Problem Definition

This PP-Module is written to address the situation where a distributed network device facilitates end user access to a network via LiFi. The threats are similar to those that are typical of network devices as a whole, but may be exploited differently because of the mechanisms that are unique to this particular type of product.

3.1 Threats

T.DATA_INTEGRITY

Devices on a protected network may be exposed to threats presented by devices located outside the protected network, which may attempt to modify the data without authorization. If known malicious external devices are able to communicate with devices on the protected network or if devices on the protected network can establish communications with those external devices, then the data contained within the communications may be susceptible to a loss of integrity.

T.NETWORK_DISCLOSURE

Devices on a protected network may be exposed to threats presented by devices located outside the protected network, which may attempt to conduct unauthorized activities. If malicious external devices are able to communicate with devices on the protected network, or if devices on the protected network can establish communications with those external devices (e.g., as a result of nonexistent or insufficient data encryption that exposes the data in transit to rogue elements), then those internal devices may be susceptible to the unauthorized disclosure of information.

T.NETWORK_ACCESS

Devices located outside the protected network may seek to exercise services located on the protected network that are intended to be only accessed from inside the protected network or only accessed by entities using an authenticated path into the protected network.

T.NETWORK_ATTACK

An attacker is positioned on a communications channel or elsewhere on the network infrastructure. Attackers may engage in communications with applications and services running on or part of the TOE with the intent of compromise. Engagement may consist of altering existing legitimate communications.

T.REPLAY_ATTACK

If an unauthorized individual successfully gains access to the system, the adversary may have the opportunity to conduct a "replay" attack. This method of attack allows the individual to capture packets traversing throughout the wireless network and send the packets later, possibly unknown by the intended receiver.

3.2 Assumptions

These assumptions are made on the Operational Environment (OE) in order to be able to ensure that the security functionality specified in the PP-Module can be provided by the TOE. If the TOE is placed in an OE that does not meet these assumptions, the TOE may no longer be able to provide all of its security functionality.

A.NON_SECURITY_FUNCTIONS

The product implements some security-relevant functionality that does not require evaluation (e.g., network time synchronization, process scheduling, and virtual memory management including process separation).

A.MANAGED_CONFIG

Depending on configuration and capability, the product may or may not be configuration-managed by the enterprise or bound to directory services to support multi-user login.

A.THIRD_PARTY_SOFTWARE

The product runs application software developed by a third party. The applications are not intentionally developed to be malicious, but can contain inadvertent coding errors. These errors introduce a risk that control of an application may be seized by a malicious entity. The product shall confine these applications within the originally designated operating environment.

A.NETWORK_CONNECTIVITY

The platform is connected to a network. For purposes of sending and receiving data, to include software updates, the platform is connected to other entities. Other entities on the network are not inherently trustable.

A.PROPER_USER

Users are not malicious in nature, though they may inadvertently or intentionally engage in risky behavior.

3.3 Organizational Security Policies

This document does not define any additional OSPs.

4 Security Objectives

4.1 Security Objectives for the Operational Environment

The OE of the TOE implements technical and procedural measures to assist the TOE in correctly providing its security functionality (which is defined by the security objectives for the TOE). The security objectives for the OE consist of a set of statements describing the goals that the OE should achieve. This section defines the security objectives that are to be addressed by the IT domain or by non-technical or procedural means. The assumptions identified in Section 3 are incorporated as security objectives for the environment.

OE.NON_SECURITY_FUNCTIONS

The product has functionality outside of its logical security boundary to satisfy organizational requirements that are unrelated to what this PP-Module defines.

OE.MANAGED_CONFIG

The product is compatible with enterprise architecture that allows for managed configuration, subject authorization via directory services, or both.

OE.THIRD_PARTY_SOFTWARE

The operational environment permits the use of third-party software on organization-owned hardware to enforce necessary functions.

OE.NETWORK_CONNECTIVITY

The operational environment facilitates the use of network connectivity by authorized subjects.

OE.PROPER_USER

Users of the system are adequately vetted to ensure that they are non-malicious.

4.2 Security Objectives Rationale

This section describes how the assumptions and organizational security policies map to operational environment security objectives.

Table 1: Security Objectives Rationale

Assumption or OSP	Security Objectives	Rationale
A.NON_SECURITY_FUNCTIONS	OE.NON_SECURITY_FUNCTIONS	The operational environment objective OE.NON_SECURITY_FUNCTIONS is realized through A.NON_SECURITY_FUNCTIONS.
A.MANAGED_CONFIG	OE.MANAGED_CONFIG	The operational environment objective OE.MANAGED_CONFIG is realized through A.MANAGED_CONFIG.
A.THIRD_PARTY_SOFTWARE	OE.THIRD_PARTY_SOFTWARE	The operational environment objective OE.THIRD_PARTY_SOFTWARE is realized through A.THIRD_PARTY_SOFTWARE.

A.NETWORK_ CONNECTIVITY	OE.NETWORK_ CONNECTIVITY	The operational environment objective OE.NETWORK_CONNECTIVITY is realized through A.NETWORK_CONNECTIVITY.
A.PROPER_USER	OE.PROPER_USER	The operational environment objective OE.PROPER_USER is realized through A.PROPER_USER.

5 Security Requirements

This chapter describes the security requirements which have to be fulfilled by the product under evaluation. Those requirements comprise functional components from Part 2 and assurance components from Part 3 of [CC]. The following conventions are used for the completion of operations:

- **Refinement** operation (denoted by **bold text** or ~~strikethrough text~~): Is used to add details to a requirement or to remove part of the requirement that is made irrelevant through the completion of another operation, and thus further restricts a requirement.
- **Selection** (denoted by *italicized text*): Is used to select one or more options provided by the [CC] in stating a requirement.
- **Assignment** operation (denoted by *italicized text*): Is used to assign a specific value to an unspecified parameter, such as the length of a password. Showing the value in square brackets indicates assignment.
- **Iteration** operation: Is indicated by appending the SFR name with a slash and unique identifier suggesting the purpose of the operation, e.g. "/EXAMPLE1."

5.1 Collaborative Protection Profile for Network Device Security Functional Requirements Direction

In a PP-Configuration that includes the NDcPP, the TOE is expected to rely on some of the security functions implemented by the network device as a whole and evaluated against the Base-PP. In this case, the following sections describe any modifications that the ST author must make to the SFRs defined in the Base-PP in addition to what is mandated by section 5.2.

5.1.1 Modified SFRs

This PP-Module does not modify any SFRs defined by the NDcPP.

5.2 TOE Security Functional Requirements

The following section describes the SFRs that must be satisfied by any TOE that claims conformance to this PP-Module. These SFRs must be claimed regardless of which PP-Configuration is used to define the TOE.

5.2.1 Auditable Events for Mandatory SFRs

Table 2: Auditable Events for Mandatory Requirements

Requirement	Auditable Events	Additional Audit Record Contents
FAU_GEN.1/LiFi	No events specified	N/A
FCS_COP.1/LiFiDataEncryption	No events specified	N/A
FIA_8021X_EXT.1	None	Provided client identity (e.g., MAC address)

FIA_UAU.6	None	Origin of the attempt (e.g., IP address)
FMT_SMF.1/LiFi	No events specified	N/A
FMT_SMR_EXT.1	No events specified	N/A
FPT_ACF_EXT.1	Unauthorized attempts to perform operations against protected data	No additional information
FPT_ASLR_EXT.1	No events specified	N/A
FPT_SBOP_EXT.1	No events specified	N/A
FPT_W^X_EXT.1	No events specified	N/A
FTA_TSE.1	Denial of a session establishment due to the session establishment mechanism	• Reason for denial • Origin of establishment attempt
FTP_ITC.1/8021X	Failed attempts to establish a trusted channel (including IEEE 802.11)	Identification of the initiator and target of channel
	Detection of modification of channel data	Identification of the initiator and target of channel

5.2.2 Security Audit (FAU)

FAU_GEN.1/LiFi Audit Data Generation (LiFi)

FAU_GEN.1.1/LiFi

The TSF shall be able to generate audit data of the following auditable events:

a. Start-up and shutdown of the audit functions
b. All auditable events for the [*not specified*] level of audit
c. [*auditable events defined in Table 2*
d. *failure of wireless sensor communication*
e. [**selection**: *auditable events defined in Table 8, no additional auditable events*]

]

Application Note: A TOE that conforms to a Protection Profile Configuration (PP-Configuration) containing this PP-Module can either be a standalone or distributed TOE as defined in the NDcPP. For distributed TOEs, the expectation for this PP-Module is that a LiFi AS is composed of a single controller and one or more access points (APs). If the TOE is a distributed system across multiple components, then they must be capable of sending audit data to the controller. If the TOE is a standalone system, audit data should be stored on the component that generates it.

The auditable events defined in Table 2, Table 8, and Table 10 are for the SFRs that are explicitly defined in this PP-Module and are intended to extend FAU_GEN.1 in the Base-PP (Table 10 is not referenced in the SFR as none of the SFRs in that section have auditable events associated with them). The events in the Auditable

Events table should be combined with those of the NDcPP in the context of a conforming Security Target. This table includes rows for all SFRs in the PP-Module. For any SFRs the ST claims, the corresponding audit events must also be claimed in the table. If the ST does not claim a particular SFR, then there is likewise no expectation that the auditable events for that SFR are required.

Per FAU_STG_EXT.1 in the Base-PP, the TOE must support transfer of the audit data to an external IT entity using a trusted channel.

FAU_GEN.1.2/LiFi

The TSF shall record within the audit data at least the following information:

a. Date and time of the event, type of event, subject identity (if applicable), and the outcome (success or failure) of the event; and
b. For each audit event type, based on the auditable event definitions of the functional components included in the PP, PP-Module, functional package, or ST [*additional information defined in auditable events tables for each auditable event, where applicable*].

5.2.3 Cryptographic Support (FCS)

FCS_COP.1/LiFiDataEncryption Cryptographic Operation (Data Encryption and Decryption for LiFi

FCS_COP.1.1/LiFiDataEncryption

The TSF shall perform [*symmetric-key encryption and decryption*] in accordance with a specified cryptographic algorithm [**selection**: *Cryptographic Algorithm*] and cryptographic key sizes [**selection**: *Cryptographic Key Sizes*] that meet the following: [**selection**: *List of Standards*]

The following table provides the allowable choices for completion of the selection operations in FCS_COP.1/LiFiDataEncryption. At minimum, AES-GCMP must be selected.

Table 3: Allowable choices for FCS_COP.1/LiFiDataEncryption

Identifier	Cryptographic Algorithm	Cryptographic Key Sizes	List of Standards
AES-GCMP	AES in Galois Counter Mode Protocol	256 bits	ISO/IEC 18033-3:2010 (Subclause 5.2), NIST SP 800-38D, IEEE 802.11ax-2021
AES-CBC	AES in CBC mode with non-repeating and unpredictable IVs	256 bits	ISO/IEC 18033-3:2010 (Subclause 5.2), ISO/IEC 10116:2017 (Clause 7)
AES-CCMP	AES in CCM mode Protocol	256 bits	ISO/IEC 18033-3:2010 (Subclause 5.2),

			NIST SP 800-38C, IEEE 802.11-2020
AES-GCM	AES in GCM mode with non-repeating IVs using [**selection**: *deterministic*, *RBG-based*], IV construction; the tag must be of length [**selection**: *96*, *104*, *112*, *120*, *128*]	256 bits	ISO/IEC 18033-3:2010 (Subclause 5.2), ISO/IEC 19772:2020 (Clause 10)

Application Note: This requirement includes cryptographic algorithms defined in the Base-PP because it applies to the cryptographic algorithms used specifically in support of LiFi communications, some of which can be used for other purposes. If any of the algorithms claimed in support of LiFi communications are used by the TSF solely for that purpose, it is not necessary to duplicate the claim in the Base-PP SFR. For example, if the TOE implements AES-CBC solely for LiFi communications, the Base-PP requirement FCS_COP.1/SKC does not need to replicate the AES-CBC claim made here. 256-bit GCMP is required in order to comply with FCS_CKM.1/WPA. AES-CCMP, if claimed, is required to use 256-bit keys for IEEE 802.11ax-2021 connections.

5.2.4 Identification and Authentication (FIA)

FIA_8021X_EXT.1 802.1X Authentication

FIA_8021X_EXT.1.1

The TSF shall conform to IEEE Standard 802.1X for a Port Access Entity (PAE) in the “Authenticator” role.

FIA_8021X_EXT.1.2

The TSF shall support communications to a RADIUS authentication server conforming to RFCs 2865, 2866, and 3579.

FIA_8021X_EXT.1.3

The TSF shall ensure that no access to its 802.1X controlled port is given to the wireless client prior to successful completion of this authentication exchange.

Application Note: This requirement covers the TOE's role as the authenticator in an 802.1X authentication exchange. If the exchange is completed successfully, the TOE will obtain the PMK from the RADIUS server and perform the four-way handshake with the wireless client (supplicant) to begin 802.11 communications.

As indicated previously, there are at least three communication paths present during the exchange; two with the TOE as an endpoint and one with the TOE acting as a transfer point only. The TOE establishes an Extensible Authentication Protocol (EAP) over Local Area Network (EAPOL) connection with the wireless client as specified in 802.1X-2007. The TOE also establishes (or has established) a RADIUS protocol connection protected either by IPsec or RadSec (TLS) with the RADIUS server. The wireless client and RADIUS server establish an EAP-TLS session (RFC 5216); in this transaction the TOE merely takes the EAP-TLS packets from its EAPOL/RADIUS endpoint and transfers them to the other endpoint. Because the specific authentication method (TLS in this case) is opaque to the TOE, there are no requirements with respect to RFC 5126 in this PP-Module. However, the base

RADIUS protocol (2865) has an update (3579) that will need to be addressed in the implementation and evaluation activities. RADIUS accounting is implemented via conformance to RFC 2866.

Additionally, RFC 5080 contains implementation issues that will need to be addressed by developers but which levy no new requirements. The point of performing 802.1X authentication is to provide access to the network (assuming the authentication was successful and that all 802.11 negotiations are performed successfully); in the terminology of 802.1X, this means the wireless client has access to the "controlled port" maintained by the TOE.

FIA_UAU.6 Re-authenticating

FIA_UAU.6.1

The TSF shall re-authenticate the user under the conditions [*when the user changes their password, [**selection**: following TSF-initiated session locking, [**assignment**: other conditions], no other conditions]*].

5.2.5 Security Management (FMT)

FMT_SMF.1/LiFi Specification of Management Functions

FMT_SMF.1.1/LiFi

The TSF shall be capable of performing the following management functions: [*configuration of the security policy for each wireless network, including:*

- *Security type*
- *Client credentials to be used for authentication*
- *[**selection**: authentication protocol, Service Set Identifier (SSID) if the SSID is broadcast, no other security policy elements]*

].

FMT_SMR_EXT.1 No Administration from Client

FMT_SMR_EXT.1.1

The TSF shall ensure that the ability to remotely administer the TOE from a wireless client shall be disabled by default.

5.2.6 Protection of the TSF (FPT)

FPT_ACF_EXT.1 Access Controls

FPT_ACF_EXT.1.1

The TSF shall implement access controls which prohibit unprivileged users from modifying:

- Kernel and its drivers and modules
- Security audit logs
- Shared libraries
- System executables
- System configuration files
- [**assignment**: *other objects*].

FPT_ACF_EXT.1.2

The TSF shall implement access controls which prohibit unprivileged users from reading:

- Security audit logs
- System-wide credential repositories
- [**assignment**: *list of other objects*].

Application Note: "Credential repositories" refer, in this case, to structures containing cryptographic keys or passwords.

FPT_ASLR_EXT.1 Address Space Layout Randomization

FPT_ASLR_EXT.1.1

The TSF shall always randomize process address space memory locations with [**selection**: *8*, *[**assignment**: number greater than 8]*] bits of entropy except for [**assignment**: *list of explicit exceptions*].

FPT_SBOP_EXT.1 Stack Buffer Overflow Protection

FPT_SBOP_EXT.1.1

The TSF shall [**selection**: *employ stack-based buffer overflow protections*, *not store parameters or variables in the same data structures as control flow values*].

Application Note: Many OSes store control flow values (i.e., return addresses) in stack data structures that also contain parameters and variables. For these OSes, it is expected that most of the OS, to include the kernel, libraries, and application software from the OS vendor be compiled with stack-based buffer overflow protection enabled. OSes that store parameters and variables separately from control flow values do not need additional stack protections.

5.2.7 TOE Access (FTA)

FTA_TSE.1 TOE Session Establishment

FTA_TSE.1.1

The TSF shall be able to deny session establishment **of a wireless client session** based on [*TOE interface, time, day, [**selection**: [**assignment**: other attributes], no other attributes]*].

Application Note: The "TOE interface" can be specified in terms of the device in the TOE that the LiFi client is connecting to (e.g., specific LiFi APs). "time" and "day" refer to time-of-day and day-of-week, respectively.

The assignment is to be used by the ST author to specify additional attributes on which denial of session establishment can be based.

5.2.8 Trusted Path/Channels (FTP)

FTP_ITC.1/8021X Inter-TSF Trusted Channel (802.1X)

FTP_ITC.1.1/8021X

The TSF shall provide a communication channel **using IEEE 802.1X and [selection: *IPsec*, *RADIUS over TLS*]** between itself and **an 802.1X authentication server** that is logically distinct from other communication channels and provides assured identification of its endpoints and protection of the channel data from modification or disclosure.

Application Note: This requirement has been iterated from its definition in the NDcPP to mandate the communications protocols and environmental components that a LiFi AS must use for 802.1X. IPsec or RADIUS over TLS (commonly known as "RadSec") is required at least for communications with the 802.1X authentication server. The requirement implies that not only are communications protected when they are initially established, but also on resumption after an outage.

The IT entity of "802.1X authentication server" is distinct from "authentication server" as specified in FTP_ITC.1 in the NDcPP because the latter may be used for administrator authentication rather than authorization of LiFi clients.

If IPsec is selected in FTP_ITC.1.1, then FCS_IPSEC_EXT.1 from the NDcPP must be claimed. If RADIUS over TLS is selected in FTP_ITC.1.1, then FCS_RADSEC_EXT.1 in this PP-Module must be claimed, as well as FCS_TLSC_EXT.1 from the Functional Package for TLS.

FTP_ITC.1.2/8021X

The TSF shall permit [*the TSF*] to initiate communication via the trusted channel.

FTP_ITC.1.3/8021X

The TSF shall initiate communication via the trusted channel for [*802.1X authentication*].

5.3 TOE Security Functional Requirements Rationale

The following rationale provides justification for each SFR for the TOE, showing that the SFRs are suitable to address the specified threats:

Table 4: SFR Rationale

Threat	Addressed by	Rationale
T.DATA_ INTEGRITY	FCS_COP.1/LiFiDataEncryption	Mitigates the threat by defining implementation of secure cryptographic integrity algorithms.
	FCS_CKM.1/WPA (implementation-dependent)	Mitigates the threat by generating a secure key used to encrypt and protect wireless traffic using WPA.
	FCS_CKM.2/DISTRIB (implementation-dependent)	Mitigates the threat by securely distributing 802.11 keys to clients so that they are not disclosed, maintaining the integrity of the data the key is used to protect.
	FCS_CKM.2/GTK (implementation-dependent)	Mitigates the threat by securely distributing group temporal keys to clients so that they are not disclosed, maintaining the integrity of the data the key is used to protect.
	FCS_CKM.2/PMK (implementation-dependent)	Mitigates the threat by securely receiving a key from an authentication server so that it is not disclosed, maintaining the integrity of the data the key is used to protect.
	FTP_ITC.1/Client (implementation-dependent)	Mitigates the threat by providing a secure channel for LiFi client traffic that protects the confidentiality and integrity of the data.

	FTP_ITC.1/Gvlc (implementation-dependent)	Mitigates the threat by providing a secure channel for LiFi client traffic that protects the confidentiality and integrity of the data.
T.NETWORK_DISCLOSURE	FCS_COP.1/LiFiDataEncryption	Mitigates the threat by defining implementation of secure cryptographic confidentiality algorithms.
	FIA_UAU.6	Mitigates the threat by re-authenticating users when specified criteria are met.
	FTA_TSE.1	Mitigates the threat by denying sessions based on undesirable characteristics.
	FTP_ITC.1/8021X	Mitigates the threat by enforcing the use of authentication to access the network.
	FCS_CKM.1/WPA (implementation-dependent)	Mitigates the threat by using a secure key generation function for WPA functionality.
	FCS_CKM.2/DISTRIB (implementation-dependent)	Mitigates the threat by securely distributing 802.11 keys to clients.
	FCS_CKM.2/GTK (implementation-dependent)	Mitigates the threat by securely distributing GTK keys to clients.
	FCS_CKM.2/PMK (implementation-dependent)	Mitigates the threat by securely receiving a key from an authentication server.
	FTP_ITC.1/Client (implementation-dependent)	Mitigates the threat by providing a secure channel for LiFi client traffic.
	FTP_ITC.1/Gvlc (implementation-dependent)	Mitigates the threat by providing a secure channel for LiFi client traffic.
	FCS_RADSEC_EXT.1 (selection-based)	Mitigates the threat by using a secure RADIUS implementation to authenticate clients.
	FCS_RADSEC_EXT.2 (selection-based)	Mitigates the threat by using a secure TLS configuration for RADIUS over TLS.
T.NETWORK_ACCESS	FAU_GEN.1/LiFi	Mitigates the threat by generating audit data that will detect access to the network.
	FIA_8021X_EXT.1	Mitigates the threat by enforcing authentication on access attempts.
	FIA_UAU.6	Mitigates the threat by re-authenticating users when specified criteria are met.
	FMT_SMF.1/LiFi	Mitigates the threat by defining management functions that must be protected.
	FMT_SMR_EXT.1	Mitigates the threat by preventing remote administration functions from being exercised from a wireless client.

	FTA_TSE.1	Mitigates the threat by denying sessions based on undesirable characteristics.
	FTP_ITC.1/8021X	Mitigates the threat by authenticating clients before allowing access to the controlled network.
	FIA_PSK_EXT.1 (implementation-dependent)	Mitigates the threat by defining the use of a PSK to authenticate clients.
	FCS_RADSEC_EXT.1 (selection-based)	Mitigates the threat by using a secure RADIUS implementation to authenticate clients.
	FCS_RADSEC_EXT.2 (selection-based)	Mitigates the threat by using a secure TLS configuration for RADIUS over TLS.
T.NETWORK_ATTACK	FAU_GEN.1/LiFi	Mitigates the threat by generating audit data that will detect TSF failures.
	FPT_ACF_EXT.1	Mitigates the threat by limiting access that could be obtained on the system as a result of successful network attack.
	FPT_ASLR_EXT.1	Mitigates the threat by preventing network attacks that rely on predictable program memory layout.
	FPT_SBOP_EXT.1	Mitigates the threat by preventing network attacks that rely on overwriting stack memory to execute arbitrary code.
	FPT_W^X_EXT.1	Mitigates the threat by preventing network attacks that rely on writing arbitrary executable code to memory and subsequently executing it.
T.REPLAY_ATTACK	FCS_COP.1/LiFiDataEncryption	Mitigates the threat by defining implementation of secure cryptographic integrity algorithms that prevent the disclosure of data that would be potentially suitable for replay attempts.
	FTP_ITC.1/8021X	Mitigates the threat by authenticating clients before allowing access to the controlled network and preventing replay of previous authentication sessions.
	FIA_UAU.6	Mitigates the threat by re-authenticating users when specified criteria are met, which can include if a specified time has passed or other indicators of stale authentication data.
	FTA_TSE.1	Mitigates the threat by denying sessions based on undesirable characteristics, which can include a detection of replay or other indicators of stale authentication data.

FCS_CKM.1/WPA (implementation-dependent)	Mitigates the threat by using a secure key generation function for WPA functionality that is not subject to replay or guessing.
FCS_CKM.2/DISTRIB (implementation-dependent)	Mitigates the threat by securely distributing 802.11 keys to clients so that they are not disclosed, maintaining confidentiality and preventing replay.
FCS_CKM.2/GTK (implementation-dependent)	Mitigates the threat by securely distributing GTK keys to clients so that they are not disclosed, maintaining confidentiality and preventing replay.
FCS_CKM.2/PMK (implementation-dependent)	Mitigates the threat by securely receiving a key from an authentication server so that it is not disclosed, maintaining confidentiality and preventing replay.
FTP_ITC.1/Client (implementation-dependent)	Mitigates the threat by providing a secure channel for LiFi client traffic that prevents the use of replayed data.
FTP_ITC.1/Gvlc (implementation-dependent)	Mitigates the threat by providing a secure channel for LiFi client traffic that prevents the use of replayed data.
FCS_RADSEC_EXT.1 (selection-based)	Mitigates the threat by using a secure RADIUS implementation to authenticate clients, which protects authentication data from replay attacks.
FCS_RADSEC_EXT.2 (selection-based)	Mitigates the threat by using a secure TLS configuration for RADIUS over TLS, which provides protection against replay and data modification.

6 Consistency Rationale

6.1 Collaborative Protection Profile for Network Device

6.1.1 Consistency of TOE Type

When this PP-Module extends the NDcPP, the TOE type for the overall TOE is still a network device. This PP-Module just defines the TOE as a specific type of network device with functional capabilities distinct to that type.

6.1.2 Consistency of Security Problem Definition

Table 5: Consistency of Security Problem Definition (NDcPP base)

PP-Module Threat, Assumption, OSP	Consistency Rationale
T.DATA_INTEGRITY	This threat is a specific type of failure that may result from successful exploitation of the T.WEAK_CRYPTOGRAPHY threat defined by the Base-PP. It is an extension of the Base-PP threat for communications that are specific to this PP-Module.
T.NETWORK_DISCLOSURE	This threat extends the security problem defined by the Base-PP to include the threat of a malicious entity in an untrusted network interacting with a protected entity in a trusted network. This is not addressed in the Base-PP because not all network devices are responsible for facilitating communications between separate networks. This threat is also consistent with the T.UNTRUSTED_COMMUNICATION_CHANNELS threat defined by the Base-PP because compromise of data in transit is one potential way this threat may be exploited.
T.NETWORK_ACCESS	This threat extends the security problem defined by the Base-PP to include the threat of a malicious entity in an untrusted network interacting with a protected entity in a trusted network. This is not addressed in the Base-PP because not all network devices are responsible for facilitating communications between separate networks.
T.NETWORK_ATTACK	This threat extends the security problem defined by the Base-PP to define a specific manifestation of T.SECURITY_FUNCTIONALITY_FAILURE.
T.REPLAY_ATTACK	This threat is a specific type of failure that may result from successful exploitation of the T.UNAUTHORIZED_ADMINISTRATOR_ACCESS and T.UNTRUSTED_COMMUNICATIONS_CHANNELS threats defined by the Base-PP. It is an extension of the Base-PP threat for communications that are specific to this PP-Module.

A.NON_SECURITY_FUNCTIONS	This is consistent with the Base-PP because the notion of exact PP conformance accepts the case where non-TSF functionality exists within the physical boundary of the TOE if there are no PP requirements to evaluate it.
A.MANAGED_CONFIG	This is consistent with A.TRUSTED_ADMINISTRATOR in the Base-PP by defining specific mechanisms by which the trusted administrator is defined.
A.THIRD_PARTY_SOFTWARE	This is consistent with the Base-PP because the existence of a trusted update mechanism allows for the possibility that the TOE may have implementation flaws in need of fixing.
A.NETWORK_CONNECTIVITY	This is consistent with the Base-PP because a network device inherently requires network connectivity and the Base-PP similarly does not treat external entities as automatically trusted (hence the use of trusted channels to authenticate them).
A.PROPER_USER	This is consistent with the expectations of A.TRUSTED_ADMINISTRATOR in the Base-PP by defining administrators as non-hostile but allowing for the possibility of unintentional error.

6.1.3 Consistency of OE Objectives

Table 6: Consistency of OE Objectives (NDcPP base)

PP-Module OE Objective	Consistency Rationale
OE.NON_SECURITY_FUNCTIONS	This is consistent with the Base-PP because the notion of exact PP conformance accepts the case where non-TSF functionality exists within the physical boundary of the TOE if there are no PP requirements to evaluate it.
OE.MANAGED_CONFIG	This is consistent with A.TRUSTED_ADMINISTRATOR in the Base-PP by defining specific mechanisms by which the trusted administrator is defined.
OE.THIRD_PARTY_SOFTWARE	This is consistent with the Base-PP because the existence of a trusted update mechanism allows for the possibility that the TOE may have implementation flaws in need of fixing.
OE.NETWORK_CONNECTIVITY	This is consistent with the Base-PP because a network device inherently requires network connectivity and the Base-PP similarly does not treat external entities as automatically trusted (hence the use of trusted channels to authenticate them).
OE.PROPER_USER	This is consistent with the expectations of A.TRUSTED_ADMINISTRATOR in the Base-PP by defining administrators as non-hostile but allowing for the possibility of unintentional error.

6.1.4 Consistency of Requirements

This PP-Module identifies several SFRs from the NDcPP that are needed to support LiFi Access System functionality. This is considered to be consistent because the functionality provided by the NDcPP is being used for its intended purpose. The rationale for why this does not conflict with the claims defined by the NDcPP are as follows:

Table 7: Consistency of Requirements (NDcPP base)

PP-Module Requirement	Consistency Rationale
Modified SFRs	
This PP-Module does not modify any requirements when the NDcPP is the base.	
Additional SFRs	
This PP-Module does not add any requirements when the NDcPP is the base.	
Mandatory SFRs	
FAU_GEN.1/LiFi	This SFR iterates the FAU_GEN.1 SFR defined in the Base-PP to define auditable events for the functionality that is specific to this PP-Module.
FCS_COP.1/LiFiDataEncryption	This SFR defines cryptographic behavior for functions outside the scope of what the Base-PP originally defined.
FIA_8021X_EXT.1	This SFR defines support for 802.1X communications, which is a logical interface that extends the scope of what the Base-PP originally defined.
FIA_UAU.6	This SFR defines support for re-authentication of wireless users, which are a type of subject beyond the scope of what the Base-PP originally defined.
FMT_SMF.1/LiFi	This SFR defines additional management functionality that is specific to the Module's product type and would therefore not be expected to be present in the Base-PP.
FMT_SMR_EXT.1	This SFR specifies that remote administration occurs over a dedicated channel, which does not conflict with the Base-PP.
FPT_ACF_EXT.1	This SFR specifies that unprivileged users are prohibited from reading or modifying certain parts of the system that could cause it to operate in an insecure state, which does not conflict with the Base-PP.
FPT_ASLR_EXT.1	This SFR enforces specific low-level protections on the software running on the TOE that does not conflict with any other security functionality. The ability of the TOE to implement ASLR measures does not prevent its code from performing any required function.
FPT_SBOP_EXT.1	This SFR enforces specific low-level protections on the software running on the TOE that does not conflict with any other security functionality. The ability of the TOE to implement stack-based overflow protection does not prevent its code from performing any required function.
FPT_W^X_EXT.1	This SFR enforces specific low-level protections on the software running on the TOE that does not conflict with any other security functionality. The ability of the TOE to prevent executable memory from also being writable does not prevent its code from performing any required function

FTA_TSE.1	This SFR applies restrictions on establishment of wireless communications, which is a logical interface that extends the scope of what the Base-PP originally defined.
FTP_ITC.1/8021X	This SFR iterates the FTP_ITC.1 SFR defined in the Base-PP to define trusted communication channels for the functionality that is specific to this PP-Module.
Optional SFRs	
This PP-Module does not define any Optional requirements.	
Objective SFRs	
This PP-Module does not define any Objective requirements.	
Implementation-dependent SFRs	
FCS_CKM.1/WPA	This SFR defines additional cryptographic functionality not defined in the Base-PP, but it implements this using the DRBG mechanism already defined in the Base-PP.
FCS_CKM.2/GTK	This SFR defines additional cryptographic functionality not defined in the Base-PP that is used for functionality outside the original scope of the Base-PP.
FCS_CKM.2/PMK	This SFR defines additional cryptographic functionality not defined in the Base-PP that is used for functionality outside the original scope of the Base-PP.
FIA_PSK_EXT.1	This SFR defines parameters for pre-shared key generation. The Base-PP supports pre-shared keys as a potential authentication method for IPsec. This PP-Module does not prevent this from being used but does define restrictions on how pre-shared keys may be generated and what constitutes an acceptable key. This may also be used for RadSec, which is outside the original scope of the Base-PP.
FTP_ITC.1/Client	This SFR iterates the FTP_ITC.1 SFR defined in the Base-PP to define trusted communication channels for the functionality that is specific to this PP-Module.
FCS_CKM.2/DISTRIB	This SFR defines an additional use for the cryptographic and self-protection mechanisms defined in the Base-PP.
FTP_ITC.1/Gvlc	This SFR iterates the FTP_ITC.1 SFR defined in the Base-PP to define trusted communication channels for the functionality that is specific to this PP-Module.
Selection-based SFRs	
FCS_RADSEC_EXT.1	This SFR defines the implementation of RadSec and the peer authentication method that it uses. This relies on the TLS requirements defined by the Base-PP and may also use the X.509v3 certificate validation methods specified in the Base-PP, depending on the selected peer authentication method.

FCS_RADSEC_EXT.2	This SFR defines the implementation of RadSec when pre-shared key authentication is used. This functionality is outside the original scope of the Base-PP, but it relies on the TLS client protocol implementation, cryptographic algorithms, and random bit generation functions defined by the Base-PP.

Appendix A - Optional SFRs

A.1 Strictly Optional Requirements

This PP-Module does not define any Strictly Optional SFRs or SARs.

A.2 Objective Requirements

This PP-Module does not define any Objective SFRs.

A.3 Implementation-dependent Requirements

A.3.1 Auditable Events for Implementation-Dependent SFRs

Table 8: Auditable Events for Implementation-dependent Requirements

Requirement	Auditable Events	Additional Audit Record Contents
FCS_CKM.1/WPA	No events specified	N/A
FCS_CKM.2/DISTRIB	No events specified	N/A
FCS_CKM.2/GTK	No events specified	N/A
FCS_CKM.2/PMK	No events specified	N/A
FIA_PSK_EXT.1	No events specified	N/A
FTP_ITC.1/Client	Failed attempts to establish a trusted channel (including IEEE 802.11)	Identification of the initiator and target of channel
	Detection of modification of channel data	Identification of the initiator and target of channel
FTP_ITC.1/Gvlc	Failed attempts to establish a trusted channel	Identification of the initiator and target of channel
	Detection of modification of channel data	Identification of the initiator and target of channel

A.3.2 SFRs for 802.11bb Implementations

FCS_CKM.1/WPA Cryptographic Key Generation (Symmetric Keys for WPA2 Connections)

This component must be included in the ST if the TOE implements any of the following features:

- ***802.11bb Support***

FCS_CKM.1.1/WPA

The TSF shall generate **symmetric** cryptographic keys in accordance with a specified cryptographic key generation algorithm [**selection**: *Cryptographic Key Generation Algorithm*] and specified cryptographic key sizes [**selection**: *Cryptographic Key Sizes*] that meet the following: [**selection**: *List of standards*]

The following table provides the allowable choices for completion of the selection operations of FCS_CKM.1/WPA. At minimum, PRF-384 must be selected.

Table 9: Allowable Choices for FCS_CKM.1/WPA

Identifier	Cryptographic Key Generation Algorithm	Cryptographic Key Sizes	List of standards
PRF-384	Pseudorandom Function using a random bit generator as specified in FCS_RBG.1 (from Base-PP)	384 bits	IEEE 802.11-2020
PRF-512	Pseudorandom Function using a random bit generator as specified in FCS_RBG.1 (from Base-PP)	512 bits	IEEE 802.11-2020
PRF-704	Pseudorandom Function using a random bit generator as specified in FCS_RBG.1 (from Base-PP)	704 bits	IEEE 802.11-2020, IEEE 802.11ax-2021

Application Note: The cryptographic key derivation algorithm required by IEEE 802.11-2020 (Section 12.7.1.2) and verified in WPA2 certification is PRF-384, which uses the HMAC-SHA-1 function and outputs a 384-bit key. The use of GCMP is defined in IEEE 802.11ax-2021 (Section 12.5.5) and requires a Key Derivation Function (KDF) based on HMAC-SHA-256 (for 128-bit symmetric keys) or HMAC-SHA-384 (for 256-bit symmetric keys). This KDF outputs 704 bits.

This requirement applies only to the keys that are generated or derived for the communications between the AP and the client once the client has been authenticated. It refers to the derivation of the Group Temporal Key (GTK), through the Random Bit Generator (RBG) specified in this PP-Module, as well as the derivation of the Pairwise Transient Key (PTK) from the Pairwise Master Key (PMK), which is done using a random value generated by the RBG specified in this PP-Module, the HMAC function as specified in this PP-Module, as well as other information. This is specified in IEEE 802.11-2020, primarily in chapter 12. FCS_RBG.1 is defined in the NDcPP.

FCS_CKM.2/GTK Cryptographic Key Distribution (GTK)

This component must be included in the ST if the TOE implements any of the following features:

- *802.11bb Support*

FCS_CKM.2.1/GTK

The TSF shall distribute **Group Temporal Key (GTK)** in accordance with a specified cryptographic key distribution method: [**selection**: *AES Key Wrap in an EAPOL-Key frame*, *AES Key Wrap with Padding in an EAPOL-Key frame*] that meets the following: [*NIST SP 800-38F, IEEE 802.11-2020 for the packet format and timing considerations*] **and does not expose the cryptographic keys**.

Application Note: This requirement applies to the GTK that is generated by the TOE for use in broadcast and multicast messages to clients to which it is connected. 802.11-2020 specifies the format for the transfer as well as the fact that it must be wrapped by the AES Key Wrap method specified in NIST SP 800-38F.

FCS_CKM.2/PMK Cryptographic Key Distribution (PMK)

This component must be included in the ST if the TOE implements any of the following features:

- *802.11bb Support*

FCS_CKM.2.1/PMK

The TSF shall **receive the 802.11 Pairwise Master Key (PMK)** in accordance with a specified cryptographic key distribution method: [*from 802.1X Authorization Server*] that meets the following: [*IEEE 802.11-2020*] **and does not expose the cryptographic keys**.

Application Note: This requirement applies to the Pairwise Master Key that is received from the RADIUS server by the TOE. The intent of this requirement is to ensure conformant TOEs implement 802.1X authentication prior to establishing secure communications with the client. The intent is that any LiFi AS product evaluated against this PP-Module will support both WPA2-Enterprise and WPA3-Enterprise at a minimum and certificate-based authentication mechanisms and therefore disallows implementations that support only pre-shared keys. Because communications with the RADIUS server are required to be performed over a protected connection, the transfer of the PMK will be protected.

FIA_PSK_EXT.1 Pre-Shared Key Composition

This component must be included in the ST if any of the following SFRs are included:

- ***FCS_RADSEC_EXT.1***

FIA_PSK_EXT.1.1

The TSF shall be able to use pre-shared keys for [**selection**: *RADIUS over TLS (RadSec)*, *IPsec*, *WPA3-SAE*, *WPA3-SAE-PK*, *IEEE 802.11 WPA2-PSK*, *[**assignment**: other protocols that use pre-shared keys]*].

FIA_PSK_EXT.1.2

The TSF shall be able to accept text-based pre-shared keys that:

- are 22 characters and [**selection**: *[**assignment**: other supported lengths]*, *no other lengths*];

- are composed of any combination of upper and lower case letters, numbers, and special characters (that include: “!”, “@”, “#”, “$”, “%”, “^”, “&”, “*”, “(“, and “)”).

FIA_PSK_EXT.1.3

The TSF shall be able to [**selection**: *accept*, *generate using the random bit generator specified in FCS_RBG.1*] bit-based pre-shared keys.

Application Note: This requirement must be included if IPsec or another protocol that uses pre-shared keys is claimed, and pre-shared key authentication is selected (e.g., "Pre-shared Keys" is selected in FCS_IPSEC_EXT.1.13 or "pre-shared keys" is selected in FCS_RADSEC_EXT.1.2). The intent of this requirement is that all protocols will support both text-based and bit-based pre-shared keys.
For the length of the text-based pre-shared keys, a common length (22 characters) is required to help promote interoperability. If other lengths are supported, they should be listed in the assignment; this assignment can also specify a range of values (e.g., "lengths from 5 to 55 characters").

For FIA_PSK_EXT.1.3, the ST author specifies whether the TSF merely accepts bit-based pre-shared keys or is capable of generating them. If it generates them, the requirement specifies that they must be generated using the RBG provided by the TOE.

FTP_ITC.1/Client Inter-TSF Trusted Channel (LiFi 802.11 Client Access)

This component must be included in the ST if the TOE implements any of the following features:

- ***802.11bb Support***

FTP_ITC.1.1/Client

The TSF shall provide a communication channel **using WPA3-Enterprise, WPA2-Enterprise, and [selection:** ***WPA3-SAE, WPA3-SAE-PK, WPA2-PSK, no other mode*****] as defined by IEEE 802.11-2020 to provide a trusted channel** between itself and **LiFi clients** that is logically distinct from other communication channels and provides assured identification of its endpoints and protection of the channel data from modification or disclosure.

Application Note: This requirement has been iterated from its definition in the NDcPP to mandate support for 802.11 specifically for LiFi client access.

FTP_ITC.1.2/Client

The TSF shall permit [***LiFi clients***] to initiate communication via the trusted channel.

FTP_ITC.1.3/Client

The TSF shall initiate communication via the trusted channel for [*no services*].

A.3.3 SFRs for Distributed TOE Implementations

FCS_CKM.2/DISTRIB Cryptographic Key Distribution (802.11 Keys)

This component must be included in the ST if the TOE implements any of the following features:

- ***Distributed TOE***

FCS_CKM.2.1/DISTRIB

The TSF shall distribute **IEEE 802.11** keys in accordance with a specified cryptographic key distribution method [**assignment**: *trusted channel protocol specified in FPT_ITT.1 (from Base-PP)*] that meets the following: [**assignment**: *applicable standards as referenced by the distribution method specified in the Base-PP*].

Application Note: This SFR is required for use in distributed TOEs that support 802.11bb for distribution of keys to access nodes. G.vlc TOEs will not claim this as they are used for point-to-point communications and do not have a distributed architecture.

If claimed, this requirement applies to any key necessary for successful IEEE 802.11 connections not covered by FCS_CKM.2/GTK. In cases where a key must be distributed to other APs, this communication must be performed via a mechanism of commensurate cryptographic strength. Because communications with any component of a distributed TOE are required to be performed over a trusted connection, the transfer of these keys will be protected.

A.3.4 SFRs for G.vlc Implementations

FTP_ITC.1/Gvlc Inter-TSF Trusted Channel (MACsec)

This component must be included in the ST if the TOE implements any of the following features:

- ***G.vlc Support***

FTP_ITC.1.1/Gvlc

The TSF shall provide a communication channel **for G.vlc using MACsec to provide a trusted channel** between itself and **a MACsec peer** that is logically distinct from other communication channels and provides assured identification of its endpoints and protection of the channel data from modification or disclosure.

Application Note: This requirement has been iterated from its definition in the NDcPP to mandate support for MACsec.

FTP_ITC.1.2/Gvlc

The TSF shall permit [**selection**: *the TSF, **a MACsec peer***] to initiate communication via the trusted channel.

FTP_ITC.1.3/Gvlc

The TSF shall initiate communication via the trusted channel for [*communications with MACsec peers that require the use of MACsec*].

Appendix B - Selection-based Requirements

B.1 Auditable Events for Selection-based SFRs

Table 10: Auditable Events for Selection-based Requirements

Requirement	Auditable Events	Additional Audit Record Contents
FCS_RADSEC_EXT.1	No events specified	N/A
FCS_RADSEC_EXT.2	No events specified	N/A

B.2 Cryptographic Support (FCS)

FCS_RADSEC_EXT.1 RadSec

The inclusion of this selection-based component depends upon selection in FTP_ITC.1.1/8021X.

FCS_RADSEC_EXT.1.1

The TSF shall implement RADIUS over TLS as specified in RFC 6614 to communicate securely with a RADIUS server.

FCS_RADSEC_EXT.1.2

The TSF shall perform peer authentication using [**selection**: *X.509v3 certificates*, *pre-shared keys*].

Application Note: This SFR is applicable if "RADIUS over TLS" is selected in FTP_ITC.1/8021X.

If X.509v3 certificates is selected in FCS_RADSEC_EXT.1.2, then FCS_TLSC_EXT.1 from the Functional Package for TLS must be claimed and "mutual authentication" must be selected in FCS_TLSC_EXT.1.1. If pre-shared keys is selected in FCS_RADSEC_EXT.1.2, then FCS_RADSEC_EXT.2 and FIA_PSK_EXT.1 in this PP-Module must be claimed.

FCS_RADSEC_EXT.2 RadSec Using Pre-Shared Keys

The inclusion of this selection-based component depends upon selection in FCS_RADSEC_EXT.1.2.

FCS_RADSEC_EXT.2.1

The TSF shall implement **TLS 1.2 and** [**selection**: *TLS 1.3*, *no other TLS versions*] when acting as a RADIUS over TLS client that supports the following ciphersuites: [**selection**:

- *TLS 1.2 ciphersuites:[**selection**:*
    - *TLS_ECDHE_PSK_WITH_AES_256_GCM_SHA384 as defined in RFC 8442*
    - *TLS_DHE_PSK_WITH_AES_256_GCM_SHA384 as defined in RFC 5487*

  ]
- *TLS 1.3 ciphersuites:*
    - *TLS_AES_256_GCM_SHA384 as defined in RFC 8446*

  *[**selection**:*
    - *TLS_AES_128_GCM_SHA256 as defined in RFC 8446*
    - *[**assignment**: other TLS 1.3 ciphersuites]*

  ]

].

Application Note: The above ciphersuites are only for use when the TSF is acting as a RADIUS over TLS client, not for other uses of the TLS protocol. The ciphersuites to be tested in the evaluated configuration are limited by this requirement. The ST author should select the ciphersuites that are supported.

This requirement must be included if pre-shared keys is selected in FCS_RADSEC_EXT.1.2.

FCS_RADSEC_EXT.2.2

The TSF shall be able to [**selection**: *accept*, *generate using the random bit generator specified in FCS_RBG.1*] bit-based pre-shared keys.

Appendix C - Extended Component Definitions

This appendix contains the definitions for all extended requirements specified in the PP-Module.

C.1 Extended Components Table

All extended components specified in the PP-Module are listed in this table:

Table 11: Extended Component Definitions

Functional Class	Functional Components
Cryptographic Support (FCS)	FCS_RADSEC_EXT RadSec
Identification and Authentication (FIA)	FIA_8021X_EXT 802.1X Port Access Entity (Authenticator) Authentication FIA_PSK_EXT Pre-Shared Keys
Protection of the TSF (FPT)	FPT_ACF_EXT Access Controls FPT_ASLR_EXT Address Space Layout Randomization FPT_SBOP_EXT Stack Buffer Overflow Protection FPT_W^X_EXT Write XOR Execute Memory Pages
Security Management (FMT)	FMT_SMR_EXT Security Management Restrictions

C.2 Extended Component Definitions

C.2.1 Cryptographic Support (FCS)

This PP-Module defines the following extended components as part of the FCS class originally defined by CC Part 2:

C.2.1.1 FCS_RADSEC_EXT RadSec

Family Behavior

Components in this family describe requirements for implementation of the RadSec (RADIUS over TLS) protocol.

Component Leveling

FCS_RADSEC_EXT.1, RadSec, requires the TSF to implement RadSec using a specified peer authentication method.

FCS_RADSEC_EXT.2, RadSec Using Pre-Shared Keys, requires the TSF to implement RadSec using pre-shared key authentication in a manner that conforms to relevant TLS specifications.

Management: FCS_RADSEC_EXT.1

No specific management functions are identified.

Audit: FCS_RADSEC_EXT.1

There are no auditable events foreseen.

FCS_RADSEC_EXT.1 RadSec

Hierarchical to:	No other components.
Dependencies to:	FCS_TLSC_EXT.1 TLS Client Protocol FIA_PSK_EXT.1 Pre-Shared Key Composition FIA_X509_EXT.1 X.509v3 Certificate Validation

FCS_RADSEC_EXT.1.1

The TSF shall implement RADIUS over TLS as specified in RFC 6614 to communicate securely with a RADIUS server.

FCS_RADSEC_EXT.1.2

The TSF shall perform peer authentication using [**assignment**: *some authentication method*].

Management: FCS_RADSEC_EXT.2

No specific management functions are identified.

Audit: FCS_RADSEC_EXT.2

There are no auditable events foreseen.

FCS_RADSEC_EXT.2 RadSec Using Pre-Shared Keys

Hierarchical to:	No other components.
Dependencies to:	FCS_CKM.1 Cryptographic Key Generation FCS_COP.1 Cryptographic Operation FCS_RADSEC_EXT.1 RadSec FCS_RBG.1 Random Bit Generation

FCS_RADSEC_EXT.2.1

The TSF shall implement [**assignment**: *list of allowed TLS versions*] when acting as a RADIUS over TLS client that supports the following ciphersuites: [**assignment**: *list of supported ciphersuites*].

FCS_RADSEC_EXT.2.2

The TSF shall be able to [**selection**: *accept*, *generate using the random bit generator specified in FCS_RBG.1*] bit-based pre-shared keys.

C.2.2 Identification and Authentication (FIA)

This PP-Module defines the following extended components as part of the FIA class originally defined by CC Part 2:

C.2.2.1 FIA_8021X_EXT 802.1X Port Access Entity (Authenticator) Authentication

Family Behavior

Components in this family describe requirements for implementation of 802.1X port-based network access control.

Component Leveling

FIA_8021X_EXT — 1

FIA_8021X_EXT.1, 802.1X Authentication, requires the TSF to securely implement IEEE 802.1X as an authenticator.

Management: FIA_8021X_EXT.1

No specific management functions are identified.

Audit: FIA_8021X_EXT.1

The following actions should be auditable if FAU_GEN Security Audit Data Generation is included in the ST:

- Attempts to access the 802.1X controlled port prior to successful completion of the authentication exchange

FIA_8021X_EXT.1 802.1X Authentication

Hierarchical to: No other components.

Dependencies to: No dependencies

FIA_8021X_EXT.1.1

The TSF shall conform to IEEE Standard 802.1X for a Port Access Entity (PAE) in the “Authenticator” role.

FIA_8021X_EXT.1.2

The TSF shall support communications to a RADIUS authentication server conforming to RFCs 2865, 2866, and 3579.

FIA_8021X_EXT.1.3

The TSF shall ensure that no access to its 802.1X controlled port is given to the wireless client prior to successful completion of this authentication exchange.

C.2.2.2 FIA_PSK_EXT Pre-Shared Keys

Family Behavior

Components in this family describe requirements for using pre-shared keys.

Component Leveling

FIA_PSK_EXT ——— 1

FIA_PSK_EXT.1, Pre-Shared Key Composition, requires the TSF to support pre-shared keys that meet various characteristics for specific communications usage.

Management: FIA_PSK_EXT.1

No specific management functions are identified.

Audit: FIA_PSK_EXT.1

There are no auditable events foreseen.

FIA_PSK_EXT.1 Pre-Shared Key Composition

Hierarchical to: No other components.

Dependencies to: FCS_RBG.1 Random Bit Generation

FIA_PSK_EXT.1.1

The TSF shall be able to use pre-shared keys for [**assignment**: *pre-shared key protocols*].

FIA_PSK_EXT.1.2

The TSF shall be able to accept text-based pre-shared keys that:

- are 22 characters and [**selection**: *[**assignment**: other supported lengths]*, *no other lengths*];
- are composed of any combination of upper and lower case letters, numbers, and special characters (that include: “!”, “@”, “#”, “$”, “%”, “^”, “&”, “*”, “(“, and “)”).

FIA_PSK_EXT.1.3

The TSF shall be able to [**selection**: *accept*, *generate using the random bit generator specified in FCS_RBG.1*] bit-based pre-shared keys.

C.2.3 Protection of the TSF (FPT)

This PP-Module defines the following extended components as part of the FPT class originally defined by CC Part 2:

C.2.3.1 FPT_ACF_EXT Access Controls

Family Behavior

This family defines specific TOE components that are protected against unprivileged access. This is a new family defined for the FPT class.

Component Leveling

FPT_ACF_EXT ——— 1

FPT_ACF_EXT.1, Access Controls, requires the TSF to prohibit unauthorized users from reading or modifying specific TSF data.

Management: FPT_ACF_EXT.1

There are no management functions foreseen.

Audit: FPT_ACF_EXT.1

The following actions should be auditable if FAU_GEN Security audit data generation is included in the PP or ST:

- Unauthorized attempts to perform operations against protected data

FPT_ACF_EXT.1 Access Controls

Hierarchical to: No other components.

Dependencies to: No dependencies.

FPT_ACF_EXT.1.1

The TSF shall implement access controls which prohibit unprivileged users from modifying:

- Kernel and its drivers and modules
- Security audit logs
- Shared libraries
- System executables
- System configuration files
- [**assignment**: *other objects*].

FPT_ACF_EXT.1.2

The TSF shall implement access controls which prohibit unprivileged users from reading:

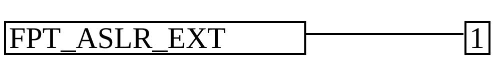
- Security audit logs
- System-wide credential repositories
- [**assignment**: *list of other objects*].

C.2.3.2 FPT_ASLR_EXT Address Space Layout Randomization

Family Behavior

This family defines the ability of the TOE to implement address space layout randomization (ASLR). This is a new family defined for the FPT class.

Component Leveling



FPT_ASLR_EXT.1, Address Space Layout Randomization, defines the ability of the TOE to use ASLR as well as the objects that ASLR is applied to.

Management: FPT_ASLR_EXT.1

There are no management functions foreseen.

Audit: FPT_ASLR_EXT.1

There are no auditable events foreseen.

FPT_ASLR_EXT.1 Address Space Layout Randomization

Hierarchical to: No other components.

Dependencies to: No dependencies.

FPT_ASLR_EXT.1.1

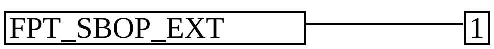
The TSF shall always randomize process address space memory locations with [**selection**: *8, [**assignment**: number greater than 8]*] bits of entropy except for [**assignment**: *list of explicit exceptions*].

C.2.3.3 FPT_SBOP_EXT Stack Buffer Overflow Protection

Family Behavior

This family requires the TSF to be compiled using stack-based buffer overflow protections. This is a new family defined for the FPT class.

Component Leveling



FPT_SBOP_EXT.1, Stack Buffer Overflow Protection, requires the TSF to be compiled using stack-based buffer overflow protections or to store data in such a manner that a stack-based buffer overflow cannot compromise the TSF.

Management: FPT_SBOP_EXT.1

There are no management activities foreseen.

Audit: FPT_SBOP_EXT.1

There are no auditable events foreseen.

FPT_SBOP_EXT.1 Stack Buffer Overflow Protection

Hierarchical to: No other components.

Dependencies to: No dependencies.

FPT_SBOP_EXT.1.1

The TSF shall [**selection**: *employ stack-based buffer overflow protections, not store parameters or variables in the same data structures as control flow values*].

C.2.3.4 FPT_W^X_EXT Write XOR Execute Memory Pages

Family Behavior

This family defines the ability of the TOE to implement data execution prevention (DEP) by preventing memory from being both writable and executable. This is a new family defined for the FPT class.

Component Leveling

FPT_W^X_EXT —— 1

FPT_W^X_EXT.1, Write XOR Execute Memory Pages, defines the ability of the TOE to prevent memory from being simultaneously writable and executable unless otherwise specified.

Management: FPT_W^X_EXT.1

There are no management functions foreseen.

Audit: FPT_W^X_EXT.1

There are no auditable events foreseen.

FPT_W^X_EXT.1 Write XOR Execute Memory Pages

Hierarchical to: No other components.

Dependencies to: No dependencies.

FPT_W^X_EXT.1.1

The TSF shall prevent allocation of any memory region with both write and execute permissions except for [**assignment**: *list of exceptions*].

C.2.4 Security Management (FMT)

This PP-Module defines the following extended components as part of the FMT class originally defined by CC Part 2:

C.2.4.1 FMT_SMR_EXT Security Management Restrictions

Family Behavior

Components in this family describe architectural restrictions on security administration that are not defined in CC Part 2.

Component Leveling

FMT_SMR_EXT —— 1

FMT_SMR_EXT.1, No Administration from Client, requires the TSF to reject remote administration from a wireless client by default.

Management: FMT_SMR_EXT.1

No specific management functions are identified.

Audit: FMT_SMR_EXT.1

There are no auditable events foreseen.

FMT_SMR_EXT.1 No Administration from Client

Hierarchical to: No other components.

Dependencies to: FMT_SMF.1 Specification of Management Functions

FMT_SMR_EXT.1.1

The TSF shall ensure that the ability to remotely administer the TOE from a wireless client shall be disabled by default.

Appendix D - Acronyms

Table 12: Acronyms

Acronym	Meaning
AES	Advanced Encryption Standard
AP	Access Point
ASLR	Address Space Layout Randomization
Base-PP	Base Protection Profile
CBC	Cipher Block Chaining
CC	Common Criteria
CCM	Counter Mode with CBC-Message Authentication Code
CCMP	CCM mode Protocol
CEM	Common Evaluation Methodology
cPP	Collaborative Protection Profile
CTR	Counter (encryption mode)
EAP	Extensible Authentication Protocol
GCM	Galois-Counter Mode
GTK	Group Temporal Key
IPsec	Internet Protocol Security
MAC	Media Access Control
NDcPP	Network Device collaborative Protection Profile
OE	Operational Environment
PAE	Port Access Entity
PMK	Pairwise Master Key
PP	Protection Profile
PP-Configuration	Protection Profile Configuration
PP-Module	Protection Profile Module

PTK	Pairwise Transient Key
RADIUS	Remote Authentication Dial In User Service
RBG	Random Bit Generator
SAR	Security Assurance Requirement
SFR	Security Functional Requirement
SSID	Service Set Identifier
ST	Security Target
TLS	Transport Layer Security
TOE	Target of Evaluation
TSF	TOE Security Functionality
TSFI	TSF Interface
TSS	TOE Summary Specification
WPA	Wi-Fi Protected Access

Appendix E - Bibliography

Table 13: Bibliography

Identifier	Title
[NDcPP]	collaborative Protection Profile for Network Devices, Version 4.0, December 22, 2025
[CC]	Common Criteria for Information Technology Security Evaluation - • Part 1: Introduction and general model, CCMB-2022-11-001, CC:2022, Revision 1, November 2022. • Part 2: Security functional requirements, CCMB-2022-11-002, CC:2022, Revision 1, November 2022. • Part 3: Security assurance requirements, CCMB-2022-11-003, CC:2022, Revision 1, November 2022. • Part 4: Framework for the specification of evaluation methods and activities, CCMB-2022-11-004, CC:2022, Revision 1, November 2022. • Part 5: Pre-defined packages of security requirements, CCMB-2022-11-005, CC:2022, Revision 1, November 2022.
[CEM]	Common Methodology for Information Technology Security Evaluation - • Evaluation methodology, CCMB-2022-11-006, CC:2022, Revision 1, November 2022.